Activity 4, Intermediate - Code Quest

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In this adventure, students will learn coding skills with CodeQuest, the free iPad app from APH. The students will program solutions to help the astronaut return to their spaceship.

Setup - Activity 4

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This activity can take several sessions. There are 30 levels within the CodeQuest app. Not all levels need to be completed. The levels start out easy and get harder as the student progresses. Students have the opportunity to work with tactile maps, 3D objects, and technology, ensuring a multi-sensory learning experience that caters to different learning styles and abilities. They are encouraged to take their time, be creative, and experiment with different solutions, fostering a sense of ownership over their code.

Objectives

- Students will be able to learn basic coding skills.
- Students will be able to understand that debugging or problem solving is part of the programming process.

Materials Needed

- iPad or iPhone
- APH CodeQuest free app (Links to App Store)
- (For braille readers) Tactical Maps of levels are very helpful while using the app
- (Optional) 3D print files for Alien, Arrows, Astronaut and Rocket Ship
- 3D printer files and grids for embossing.

Ideas for Carrying Out This Activity

- Focus on iPad skills throughout this adventure
  - Tapping

  - Swiping
  - Scanning
  - Selecting
  - VoiceOver

Adventure Map: Activity 4

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Using embossed Grid 1, challenge students to create "code" to move the astronaut to the rocket ship using the following commands. They must use each command at least once!
  - Left, Up, Down, Right
  - Consider using a small, 3D object for the students to move around the grid for trial and error.
- Any combination that results with the arrival of the astronaut to the rocket ship is acceptable.
- Next, have the student write down the commands (in order) that would be used to tell the astronaut how to follow a path to get to the rocket ship.
- If time, students can check their work by reading code out loud and having someone else follow commands to be sure that the code does not have any "bugs." Edit as necessary.

2. Kratos (1st Planet, 6 levels)

- Introduce the student to the symbols used for the maps, either on the screen or using the tactile maps. If using the tactile maps, be sure the student takes time to look at both the key and the map.
- Have the student share how they think the astronaut will have to move to get to the rocket ship. If the student has difficulty keeping track of the steps, it may help to have the student write down the steps first.
- Introduce the student to the buttons at the bottom of the screen.
- Have the student select the correct steps and then select the Play button to have the astronaut perform the steps.
- Encourage the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
- Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.

- After the student has completed all 6 levels, they can select Planet Helios to go to the next Planet.

3. **Helios (2nd Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The alien is new in this planet.
   - Let the student know that they only have to land on the square with the alien to pick it up, but they must go to each square with an alien before arriving at the rocket ship.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Attis to go to the next Planet.

4. **Attis (3rd Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The breakable wall is new in this level.
   - Let the student know that they cannot move to the square with the breakable wall until they go to a square next to the breakable wall and then use the blaster to remove it.
   - Have the student find the new Blaster button at the bottom.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Castor to go to the next Planet.

5. **Castor (4th Planet, 6 levels)**
   - Ask the student what symbol on the map is new to them. The stronger breakable wall is new in this level.
   - Let the student know that they will need to look for the number on the stronger breakable wall to find out how many times they will need to use the blaster to remove it.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.
   - After the student has completed all 6 levels, they can select Planet Pollux to go to the final Planet.

6. **Pollux (5th Planet, 6 levels)**
   - Have the student find the new Loop button at the bottom.
   - Show the student how they can select the Loop button and then swipe up or down to change the number of times to loop the command.
   - Placing this Loop command after another command makes that command repeat for the designated number of times.
   - Challenge the student to use the least number of steps possible while still getting the astronaut to the rocket ship. The Loop will help decrease the number of steps.
   - Once the student gets the astronaut to the rocket ship, they can select Next Level to go to the next map.

7. **Conclusion**
   - Summarize the key points of the adventure.
   - Have the student reflect on what went well and what was challenging.

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